

BDIC – BE IoT Project No. 5

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Programme:	BE IoT
No. Students:	4
Project Title:	A Federated Intrusion Detection Mechanism with Adaptive Privacy for UAV Swarms

Project Description:

Unmanned Aerial Vehicles (UAVs), commonly known as drones, are increasingly deployed across application domains such as urban surveillance, emergency response, environmental monitoring, and precision agriculture. Their ability to operate autonomously, collect real-time data, and collaborate as swarms makes them highly effective in complex and distributed environments. However, these capabilities also introduce significant security and privacy challenges that must be addressed at the system design level.

UAV systems are vulnerable to a range of cyber threats, including GPS spoofing, signal jamming, adversarial interference, and coordinated attacks targeting swarm communication protocols. Furthermore, drone operations over civilian and sensitive areas must comply with data protection and privacy regulations (e.g., GDPR, Ireland's Data Protection Acts, and China's PIPL), particularly when collecting telemetry, sensor readings, or image data. These constraints make traditional, centralized security solutions impractical or non-compliant.

Conventional **Intrusion** Detection Systems (IDS) typically rely on centralized machine learning models, which are unsuitable for UAV swarms due to intermittent connectivity, limited onboard computation, latency constraints, and the sensitivity of raw data. To address these limitations, Federated Learning (FL) enables collaborative model training across drones without requiring raw data to be shared. While promising, FL in UAV swarms introduces additional design challenges, including client dropout, heterogeneous hardware capabilities, malicious or unreliable participants, and the need for adaptive privacy guarantees.

In this project, students are expected to design and implement a federated IDS framework for UAV swarms that integrates Federated Learning with Reinforcement Learning (RL)-based trust modelling. The system should include an Adaptive Privacy Controller (APC) capable of dynamically adjusting privacy parameters (e.g., participation level, update frequency, or noise injection) based on real-time contextual factors such as threat level, mission criticality, trust scores, and resource availability.

Related work:

Wang, Xiaojie, Zhonghui Zhao, Ling Yi, Zhaolong Ning, Lei Guo, F. Richard Yu, and Song Guo. "A Survey on Security of UAV Swarm Networks: Attacks and Countermeasures." *ACM Computing Surveys* 57, no. 3 (2024): 1-37.

Husnoo, Muhammad Akbar, Adnan Anwar, Ripon K. Chakraborty, Robin Doss, and Mike J. Ryan. "Differential privacy for IoT-enabled critical infrastructure: A comprehensive survey." *IEEE access* 9 (2021): 153276-153304.